INCH-POUND

MIL-I-46912 (AR)
AMENDMENT 5
19 October 1990
SUPERSEDING
AMENDMENT 4
10 March 1984

MILITARY SPECIFICATION

INCREMENT, PROPELLANT, XM90 (M90)

This Amendment forms a part of Military Specification MIL-I-46912 (AR), dated 19 November 1962 and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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TITLE: Add the following preamble underneath the title:

"This specification is approved by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense."

2.1 Issues of Documents

SPECIFICATIONS:

FEDERAL: Delete "QQ-S-781"

MILITARY:

Delete "MIL-L-10547"
Delete "MIL-B-13239"
Add "MIL-B-117 - Bag, Sleeve and Tubing-Interior Packaging"

STANDARDS:

Add "MIL-STD-652 - Propellant, Solid for Cannon Requirements and Packaging"
Add "MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes"

1SC N/A FSC 1375

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DRAWINGS:

Delete "20-4-613, 76-4-59, 76-4-46, 76-4-56, and 7549033"
Add "8796522"
Delete "XM113501, XM113502, XM113503, XM113504 and substitute "8881023, 8881024, 8881021, and 8881022"

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4.2.2, line 4: Delete "Handbook ORD-M608-11" and substitute "Standard MIL-STD-1235"

4.2.2.1, line 1: Delete "113501, 113502, 113503 and 113504" and substitute "8881023, 8881024, 8881021, and 8881022"

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4.2.3.1, line 2: Delete "113501 and 113503" and substitute "8881023 and 8881021"

* 5.1.1: Delete in its entirety and substitute the following:

"5.1.1 Level A. Propellant increments shall be packed in heat sealed barrier bags complying with Type I, Class E, of MIL-B-117. The barrier bag size shall be 22 1/2 inches long by 22 1/2 inches wide, nominal. Approximately 15 pounds of propellant shall be packed in each bag and closed by means of heat sealing. Two heat sealed bags shall be packed in a wooden box (Suggested inside box dimensions 14 1/2" by 14 1/2" by 7 1/4") conforming to Grade A, Type I, Class 1 or 3 of MIL-B-2427. Both ends, sides, top, and bottom of the wooden box shall be lined with 1/16 inch thick solid water resistance sheet material conforming to MIL-F-50499. The box shall be closed and strapped in accordance with MIL-B-2427. The net weight shall not exceed 30 pounds of propellant."

* 5.1.1.1: Add new paragraph as follows:

"5.1.1.1 Alternative Level A. Propellant increments shall be packed in propellant packing boxes M17, M18, M24 or M25 complying with dwg. 76-4-56, 76-4-59, 76-4-46, and 7549033 respectively. In lieu of air pressure test, two 4-MIL conductive polyolefin or polyethylene bags can be used for all containers listed herein. Each bag must be individually tied and shall be a lay flat style bag made of conductive material in accordance with 5.1.2.1."

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* 5.1.2: Delete in its entirety and substitute the following:

"5.1.1 Level B. Propellant increments shall be packed in a fiberboard carton conforming to requirements referenced on drawing no. 7548645. The liner shall be a lay flat style bag made of conductive material in accordance with paragraph 5.1.2.1."

* 5.1.2.1, add new paragraph as follows:

"5.1.2.1 Liners. Liners shall be made of conductive polyethylene or polyolefin sheet with a thickness of .004 inch minimum, surface resistivity shall not exceed 100 ohms per centimeter, when tested in accordance with Methods 4041 and 5052 of FED-STD-406. Electrostatic bleed-off shall not exceed 2.00 seconds when tested in accordance with Method 4046 of FED-STD-101. The liner shall be closed by rolling down the top and taped (with pressure tape made in accordance with PPP-T-97 or MIL-T-43036) to preven unrolling or it may be heat sealed. Details not specified herein shall comply with PPP-B-26."

* 5.1.3: Delete in its entirety.

* 5.1.3.1: Delete in its entirety.

5.2.1: Delete in its entirety and substitute the following:

"5.2.1 Level A: Exterior containers shall be marked in accordance with drawing 8796522. The DOT marking shall be "Propellant Explosive - Class A."

* 5.2.2: Delete in its entirety and substitute the following:

"5.2.2 Alternative Level A. The M17, M18, and M24 Containers shall be marked in accordance with dwg. 8858848. The M25 Container shall be marked in accordance with dwg. 8858577."

5.2.3: Delete in its entirety and substitute the following:

"5.2.3 Level B. The fiberboard carton or other container shall be marked in accordance with dwg. 9241693."

6.2: Delete in its entirety and substitute the following:

"6.2 Inspection Equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment List (IEL) in accordance with instructions of 6.2.1 through 6.2.7."

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6.2.1: Delete in its entirety and substitute the following:

"6.2.1 Inspection Equipment Lists. Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.2.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.2.3 and 6.2.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.2.5. The contractor will be required to prepare detailed drawings in accordance with 6.2.6 for all the equipment coded as "contractor design" in the number column. These contractordesigns must be approved by the government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.2.7."

6.2.2: Delete in its entirety and substitute the following:

"6.2.2 Inspection Equipment List codes. The inspection equipment as defined in 6.2.3, 6.2.4, 6.2.5, and 6.2.6 will be designated in the Inspection Equipment List by the following codes:

CDOF - Contractor's design responsibility on Ordnance format: ORD-608-12.
CDCF - Contractor's design responsibility on contractor format.
OD - Ordnance design.
ODMU - Ordnance design, mandatory for use.
CE - Commercial equipment.
SCD - Specification Control Drawing."

6.2.3: Delete in its entirety and substitute the following:

"6.2.3 Ordnance design. Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an Inspection Equipment List references mandatory ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the government, design alternate inspection equipment or use comparable commercial

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equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.2.7."

6.2.4: Delete in its entirety and substitute the following:

"6.2.4 Specification control drawings. Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractor must prepare detailed drawings (see 6.2.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the government. Designs shall be submitted for approval as specified in 6.2.7."

6.2.5: Delete in its entirety and substitute the following:

"6.2.5 Commercial equipment. Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the government. Lists shall be submitted for approval to the inspection element of the agency administering the contract."

6.2.6: Add new paragraph to read as follows:

"6.2.6 Contractor designs. Contractor designs are designs of inspection equipment for which the government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDDF may be developed on the format the contractor normally employs in his equipment

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design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608-12, and in addition, contain the detail and information specified above."

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6.2.7: Add new paragraph to read as follows:

"6.2.7 Submission of contractor designs. Designs shall be submitted for approval to Commander, U.S. Army ARRADCOM, ATTN: DRDAR-QAT-I, Picatinny Arsenal, NJ 07806-5000. Design review will normally be accomplished within one month after receipt by ARRADCOM. Partial submission of inspection equipment designs is permissible and encouraged. However, the ARRADCOM completion date for design review will be based on the date of the final submission of designs."

NOTE: The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Preparing activity:
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(Project 1375-A454)

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